

Ethan Gates

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SUMMARY

AI-Native Software Engineer building production AI products, agent systems, and data-intensive full-stack applications. Healthcare SaaS engineer with hands-on work in grounded AI, local inference, orchestration, and evaluation.

EDUCATION

B.S. Computer Science, Western Governors University Completed: Feb 2025

PROFESSIONAL EXPERIENCE

Software Engineer, DOVAXIS (Healthcare SaaS patient referrals & admissions) May 2025 – Present
Began as a paid engineering intern; converted to full-time November 2025.

- Built governed document-intelligence workflows with lifecycle/version controls, curated regulation search, grounded AI answers with citations, retrieval debugging, and usage/cost analytics.
- Consolidated scattered role and resource checks into a Permit.io-backed authorization facade with bulk decisions and cached permission loading across routes, navigation, folders, documents, eForms, and regulations.
- Hardened a legacy React/Firebase codebase without a freeze: turned on type checking and ESLint in CI behind shrink-only baselines, cut TypeScript errors 29%, removed 113k lines of unreachable code, and ran Jest adapters and Firestore rules against the real Firebase engine on every pull request.
- Built test-governance and Firebase-emulator workflows that move data-writing checks off production, enforce suite execution policies, and surface static-check evidence in pull-request scorecards.
- Delivered 479 merged pull requests in the core healthcare SaaS repository from May 2025 through September 2026; in the 90 days to September 4 authored 227 of the 342 commits that landed on main (66%, next contributor 19%).

INDEPENDENT PROJECTS

- **GatesAI Chat**: Built and shipped a local-first AI workspace (React/Tauri/Rust with a Go bridge) featuring OpenRouter/Ollama routing, multi-round tool execution, sandboxed files/shell/Git/SQLite, memory, MCP, and image generation, with automated checks for provider streams, tools, persistence, and user journeys. [\[Repo\]](#)
- **Fathom**: Built a market-analysis suite with 155 years of asset-class history, portfolio backtesting, Monte Carlo retirement simulation, SEC-filing research, and broker-CSV portfolio X-ray as a pure-TypeScript analytics engine with a React frontend. [\[Demo\]](#)
- **N2K Almanac**: Engineered a TypeScript solver computing the optimal equation for 370K+ dice/target combinations, encoded in a custom compact binary format and served through a Web Worker for instant lookups, with a React/MobX data-visualization frontend. [\[Demo\]](#)
- **Neon Vector Defense**: Shipped a hosted React/TypeScript Canvas tower-defense game with deterministic simulation, bot playtests, headless balance harnesses, compact replay verification, Firebase leaderboards, and server-side anti-cheat. [\[Demo\]](#)

TECHNICAL SKILLS

- **Languages**: TypeScript, Python, Rust, JavaScript, SQL, C#.
- **AI / LLM Engineering**: Agent orchestration, tool calling, grounded retrieval/citations, embeddings, evaluation & benchmarking, OpenRouter, Ollama, MCP.
- **Frameworks & Libraries**: React, MobX, FastAPI, Tauri, TensorFlow, NumPy, Pandas, D3.js, Tailwind CSS.
- **Tools & Infrastructure**: Git/GitHub, Playwright, Firebase, PostgreSQL, MySQL, Docker, Vite, Google Cloud Run, Linux.